

A Generating-Function Framework for the HK-Property in Perfect m -ary Trees

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Abstract

We develop a novel generating function framework to study the HK-property in perfect m -ary trees. By encoding the distribution of independent sets through generating functions, we establish a coefficientwise comparison between independent sets containing a fixed leaf and those containing the root. This approach yields a simultaneous induction proof of the HK-property for all perfect m -ary trees, along with a structured recurrence for the difference between leaf and root stars. Our method provides new insights into the combinatorial structure of independent sets in trees and opens avenues for further exploration of EKR-type phenomena.

1 Introduction

The Erdős–Ko–Rado theorem [2] is a foundational result about intersecting families of uniform subsets. One graph-theoretic version studies intersecting families of independent sets: for a graph G , one asks whether the largest intersecting subfamilies of independent k -sets are obtained by fixing a vertex. Such fixed-vertex families are called stars. A graph satisfying this extremal property is called k -EKR.

Hurlbert and Kamat [5] proposed a stronger star-centre phenomenon for trees: in any tree, the largest stars should be centred at leaves. This statement, often called the HK-property, is false for general trees, but it remains true for several natural classes. In particular, Ľkovič and Yan [6] recently proved the HK-property for all perfect m -ary trees by a combinatorial injection.

This paper develops a complementary approach. Instead of constructing injections between individual star families, we encode the entire distribution of independent sets by generating functions. For the perfect m -ary tree $T_{m,n}$ of depth n , we introduce three polynomials:

$$F_n(x), \quad L_n(x), \quad R_n(x),$$

counting all independent sets, independent sets containing a fixed leaf, and independent sets containing the root, respectively. These polynomials satisfy a coupled recursive system. The HK comparison between leaves and the root then becomes the coefficientwise inequality

$$L_n(x) \geq_c R_n(x).$$

The main contribution is a coefficientwise proof of this inequality for all $m \geq 2$ and all depths n . The proof proceeds by a simultaneous induction driven by two alternating defect sequences. This gives a generating-function proof of the HK-property for perfect m -ary trees.

The same framework also quantifies the gap. Writing

$$D_n(x) := L_n(x) - R_n(x),$$

we derive a recursive decomposition for D_n with an explicitly controlled correction term. Thus the HK-property is not only recovered as a non-negativity statement; the difference between the leaf star and root star is itself shown to satisfy a structured recurrence.

Finally, we record several EKR consequences and related phenomena for perfect m -ary trees. Some of these follow naturally from the structure of the tree, such as the formulas for the independence number and for the minimum size of a maximal independent set. Other results, including low-uniformity classifications and top-layer failures, are logically independent of the generating-function proof and are collected separately.

2 Preliminaries and Notation

Throughout the paper, $m \geq 2$ is fixed. Let $T_{m,n}$ denote the perfect m -ary tree of depth n : the root has depth 0, every non-leaf vertex has exactly m children, and all leaves have depth n . Thus

$$N_{m,n} := |V(T_{m,n})| = 1 + m + \cdots + m^n = \frac{m^{n+1} - 1}{m - 1}.$$

The maximum independent set is the parity class

$$A_{m,n} = \{\text{vertices at depths } n, n-2, n-4, \dots\},$$

and hence

$$\alpha_{m,n} := \alpha(T_{m,n}) = m^n + m^{n-2} + m^{n-4} + \cdots.$$

For $n \geq 1$, the minimum size of a maximal independent set is

$$\mu(T_{m,n}) = m^{n-1} + m^{n-3} + m^{n-5} + \cdots = \alpha(T_{m,n-1}).$$

We use r for the root of $T_{m,n}$ and ℓ for a fixed leaf. By symmetry, the size of a leaf star does not depend on the chosen leaf.

For a graph G , let $V(G)$ and $E(G)$ denote its vertex and edge sets. An *independent set* is a subset of vertices containing no adjacent pair. We define:

$$\mathcal{I}^k(G) = \{I \subseteq V(G) : I \text{ is independent and } |I| = k\}.$$

For $v \in V(G)$, the k -star centred at v is

$$\mathcal{I}_v^k(G) = \{I \in \mathcal{I}^k(G) : v \in I\}.$$

The independence number of G is denoted by $\alpha(G)$.

A family $\mathcal{F} \subseteq \mathcal{I}^k(G)$ is *intersecting* if $A \cap B \neq \emptyset$ for all $A, B \in \mathcal{F}$. The graph G is called *k -EKR* if every intersecting family $\mathcal{F} \subseteq \mathcal{I}^k(G)$ satisfies

$$|\mathcal{F}| \leq \max_{v \in V(G)} |\mathcal{I}_v^k(G)|.$$

It is called *strictly k -EKR* if every maximum-size intersecting family is a star.

We use the coefficientwise partial order on $\mathbb{Z}[x]$. For polynomials $P, Q \in \mathbb{Z}[x]$, we write:

$$P \geq_c Q \iff P - Q \in \mathbb{N}[x].$$

Equivalently, every coefficient of $P - Q$ is non-negative. This order is preserved under addition and under multiplication by polynomials in $\mathbb{N}[x]$. We shall repeatedly use the elementary fact that if $U, V \in \mathbb{N}[x]$ and $U \geq_c V$, then

$$U^a \geq_c V^a \quad (a \geq 1),$$

because

$$U^a - V^a = (U - V) \sum_{j=0}^{a-1} U^{a-1-j} V^j.$$

The HK-property for perfect m -ary trees may be stated as follows.

Definition 2.1 (HK-property). A tree T has the *HK-property* if, for every $k \geq 1$, some leaf ℓ of T satisfies

$$|\mathcal{I}_v^k(T)| \leq |\mathcal{I}_\ell^k(T)| \quad \text{for every } v \in V(T).$$

Because $T_{m,n}$ is symmetric, it is enough to compare a leaf with the root in every rooted perfect subtree.

Lemma 2.2 (Leaf-root domination implies the HK-property). *Assume that for every $a \geq 0$ and every $k \geq 1$, the perfect m -ary tree $T_{m,a}$ satisfies*

$$|\mathcal{I}_r^k(T_{m,a})| \leq |\mathcal{I}_\ell^k(T_{m,a})|,$$

where r is its root and ℓ is any leaf. Then every perfect m -ary tree has the HK-property.

Proof. Let $T = T_{m,n}$ and let $v \in V(T)$. If v is a leaf, there is nothing to prove. Otherwise, let T_v be the rooted perfect subtree below v , and choose a descendant leaf ℓ_v of v . The subtree T_v is isomorphic to $T_{m,a}$ for some $a \geq 0$.

We compare k -sets containing v with k -sets containing ℓ_v by conditioning on the part outside T_v . Let $J \subseteq V(T) \setminus V(T_v)$ be an independent set. If J is compatible with some independent set of T_v containing v , then J does not contain the parent of v . Since the only edge between T_v and the rest of T is the edge from v to its parent, the same set J is compatible with every independent set of T_v containing ℓ_v .

For each s , the hypothesis applied inside T_v gives

$$|\mathcal{I}_v^s(T_v)| \leq |\mathcal{I}_{\ell_v}^s(T_v)|.$$

Multiplying this inequality by the number of possible outside choices J of size $k - s$ and summing over s gives

$$|\mathcal{I}_v^k(T)| \leq |\mathcal{I}_{\ell_v}^k(T)|.$$

Finally, all leaves of $T_{m,n}$ are equivalent under automorphisms. Therefore the star at ℓ_v has the same size as the star at any fixed leaf ℓ , proving the HK-property. $\square$

3 Generating functions

We now introduce the generating functions used throughout the paper. To make the recurrences uniform, we use the convention that $T_{m,-1}$ is the empty tree. Thus

$$F_{-1}(x) = 1, \quad L_{-1}(x) = 0.$$

For $n \geq 0$, define

$$F_n(x) = \sum_{k \geq 0} |\mathcal{I}^k(T_{m,n})| x^k,$$

$$L_n(x) = \sum_{k \geq 0} |\mathcal{I}_\ell^k(T_{m,n})| x^k,$$

and

$$R_n(x) = \sum_{k \geq 0} |\mathcal{I}_r^k(T_{m,n})| x^k.$$

The base values are

$$F_0(x) = 1 + x, \quad L_0(x) = R_0(x) = x.$$

Proposition 3.1 (Generating-function recurrences). *For every $n \geq 1$,*

$$F_n = F_{n-1}^m + x F_{n-2}^{m^2}, \tag{1}$$

$$R_n = x F_{n-2}^{m^2}, \tag{2}$$

$$L_n = L_{n-1} F_{n-1}^{m-1} + x L_{n-2} F_{n-2}^{m^2-1}. \tag{3}$$

Proof. For F_n , condition on whether the root is excluded or included. If the root is excluded, the m child subtrees are independent copies of $T_{m,n-1}$, contributing F_{n-1}^m . If the root is included, its m children are excluded, and the remaining descendants split into m^2 copies of $T_{m,n-2}$. The root contributes a factor of x , giving $x F_{n-2}^{m^2}$.

The formula for R_n is exactly the root-included case.

For L_n , fix a leaf ℓ . If the root is excluded, the child subtree containing ℓ contributes L_{n-1} and the other $m-1$ child subtrees contribute F_{n-1}^{m-1} . If the root is included, all children are excluded; among the m^2 grandchild subtrees, one contains ℓ and contributes L_{n-2} , while the remaining m^2-1 contribute $F_{n-2}^{m^2-1}$. The root again contributes x . $\square$

The coefficientwise HK comparison between a leaf and the root is therefore equivalent to

$$L_n \geq_c R_n.$$

We write

$$D_n := L_n - R_n \quad (n \geq 0)$$

for the HK gap polynomial. Notice that $D_0 = 0$.

4 A coefficientwise proof of the HK-property

The proof uses two alternating coefficientwise defects. These defects are the mechanism that makes the simultaneous induction work.

4.1 Two alternating defects

Lemma 4.1 (First alternating defect). *For $n \geq 2$, define*

$$A_n := F_{n-3}^{m(m-1)} F_{n-1}^{m-1} - F_{n-2}^{m^2-1}.$$

Then

$$A_n \geq_c 0 \quad \text{for odd } n \geq 3,$$

and

$$-A_n \geq_c 0 \quad \text{for even } n \geq 2.$$

Proof. Define

$$C_n := F_{n-3}^m F_{n-1} - F_{n-2}^{m+1} \quad (n \geq 2).$$

Then

$$\begin{aligned} A_n &= (F_{n-3}^m F_{n-1})^{m-1} - (F_{n-2}^{m+1})^{m-1} \\ &= C_n \sum_{j=0}^{m-2} (F_{n-3}^m F_{n-1})^{m-2-j} (F_{n-2}^{m+1})^j. \end{aligned}$$

Thus A_n has the same coefficientwise sign as C_n .

Using (1), for $n \geq 2$ we have

$$\begin{aligned} C_{n+1} &= F_{n-2}^m F_n - F_{n-1}^{m+1} \\ &= F_{n-2}^m (F_{n-1}^m + x F_{n-2}^{m^2}) - F_{n-1}^{m+1} \\ &= x F_{n-2}^{m(m+1)} - x F_{n-3}^{m^2} F_{n-1}^m \\ &= -x [(F_{n-3}^m F_{n-1})^m - (F_{n-2}^{m+1})^m] \\ &= -x C_n \sum_{j=0}^{m-1} (F_{n-3}^m F_{n-1})^{m-1-j} (F_{n-2}^{m+1})^j. \end{aligned}$$

Moreover,

$$C_2 = F_{-1}^m F_1 - F_0^{m+1} = F_1 - F_0^{m+1} = -x(F_0^m - 1) \leq_c 0.$$

Hence the coefficientwise sign of C_n alternates with n . Therefore $C_n \geq_c 0$ for odd n and $-C_n \geq_c 0$ for even n . The same sign pattern holds for A_n . $\square$

Lemma 4.2 (Second alternating defect). *For $k \geq 0$, define*

$$J_k := F_k F_{k+1}^m - F_{k-1}^m F_{k+2}.$$

Then

$$J_k \geq_c 0 \quad \text{for even } k,$$

and

$$-J_k \geq_c 0 \quad \text{for odd } k.$$

Proof. For $k \geq 0$, using (1),

$$\begin{aligned} J_{k+1} &= F_{k+1} F_{k+2}^m - F_k^m F_{k+3} \\ &= F_{k+1} F_{k+2}^m - F_k^m (F_{k+2}^m + x F_{k+1}^{m^2}) \\ &= x F_{k-1}^{m^2} F_{k+2}^m - x F_k^m F_{k+1}^{m^2} \\ &= -x [(F_k F_{k+1}^m)^m - (F_{k-1}^m F_{k+2})^m] \\ &= -x J_k \sum_{j=0}^{m-1} (F_k F_{k+1}^m)^{m-1-j} (F_{k-1}^m F_{k+2})^j. \end{aligned}$$

The base case is

$$\begin{aligned} J_0 &= F_0 F_1^m - F_{-1}^m F_2 \\ &= F_0 F_1^m - (F_1^m + x F_0^{m^2}) \\ &= x(F_1^m - F_0^{m^2}) \geq_c 0, \end{aligned}$$

because $F_1 = F_0^m + x \geq_c F_0^m$. The recurrence for J_{k+1} flips the sign at every step, proving the claim. $\square$

4.2 The main defect recurrence

Lemma 4.3 (Defect recurrence). *For every $n \geq 3$,*

$$D_n = F_{n-1}^{m-1} D_{n-1} + x F_{n-2}^{m^2-1} D_{n-2} + x F_{n-3}^m A_n. \quad (4)$$

Proof. Using (3), (2), and the definition of D_n ,

$$\begin{aligned} D_n &= L_n - x F_{n-2}^{m^2} \\ &= L_{n-1} F_{n-1}^{m-1} + x L_{n-2} F_{n-2}^{m^2-1} - x F_{n-2}^{m^2}. \end{aligned}$$

For $n \geq 3$,

$$L_{n-1} = D_{n-1} + x F_{n-3}^{m^2}, \quad L_{n-2} = D_{n-2} + x F_{n-4}^{m^2}.$$

Substituting gives

$$\begin{aligned} D_n &= F_{n-1}^{m-1} D_{n-1} + x F_{n-2}^{m^2-1} D_{n-2} \\ &\quad + x \left[F_{n-3}^{m^2} F_{n-1}^{m-1} + x F_{n-4}^{m^2} F_{n-2}^{m^2-1} - F_{n-2}^{m^2} \right]. \end{aligned}$$

Since $F_{n-2} = F_{n-3}^m + x F_{n-4}^{m^2}$, the bracket simplifies to

$$F_{n-3}^{m^2} F_{n-1}^{m-1} - F_{n-3}^m F_{n-2}^{m^2-1} = F_{n-3}^m A_n.$$

This proves (4). $\square$

We also need two comparison inequalities.

Lemma 4.4 (Even-depth comparison). *For every even $n \geq 4$,*

$$F_{n-1}^{m-1} A_{n-1} + F_{n-3}^{m-1} A_n \geq_c 0.$$

Proof. Expanding and cancelling gives

$$\begin{aligned} &F_{n-1}^{m-1} A_{n-1} + F_{n-3}^{m-1} A_n \\ &= F_{n-2}^{m-1} \left[(F_{n-4}^m F_{n-1})^{m-1} - (F_{n-3} F_{n-2}^m)^{m-1} \right]. \end{aligned}$$

Since n is even, $n-3$ is odd. By Lemma 4.2,

$$J_{n-3} = F_{n-3} F_{n-2}^m - F_{n-4}^m F_{n-1} \leq_c 0.$$

Hence $F_{n-4}^m F_{n-1} \geq_c F_{n-3} F_{n-2}^m$. Taking $(m-1)$ st powers and multiplying by F_{n-2}^{m-1} proves the result. $\square$

Lemma 4.5 (Odd-depth comparison). *For every odd $n \geq 5$,*

$$F_{n-2}^{m^2-1}A_{n-2} - F_{n-4}^{m^2-1}A_n \geq_c 0.$$

Proof. Expanding and cancelling gives

$$\begin{aligned} & F_{n-2}^{m^2-1}A_{n-2} - F_{n-4}^{m^2-1}A_n \\ &= (F_{n-5}^m F_{n-3} F_{n-2}^{m+1})^{m-1} - (F_{n-3}^m F_{n-1} F_{n-4}^{m+1})^{m-1}. \end{aligned}$$

It is enough to compare the two bases. Factor them as

$$F_{n-5}^m F_{n-3} F_{n-2}^{m+1} = (F_{n-5}^m F_{n-2})(F_{n-3} F_{n-2}^m),$$

and

$$F_{n-3}^m F_{n-1} F_{n-4}^{m+1} = (F_{n-4} F_{n-3}^m)(F_{n-4}^m F_{n-1}).$$

Since n is odd, $n-4$ is odd and $n-3$ is even. By [Lemma 4.2](#),

$$F_{n-5}^m F_{n-2} \geq_c F_{n-4} F_{n-3}^m$$

and

$$F_{n-3} F_{n-2}^m \geq_c F_{n-4}^m F_{n-1}.$$

Multiplying these two inequalities compares the bases. Taking $(m-1)$ st powers gives the lemma. $\square$

4.3 The simultaneous induction

Theorem 4.6 (Coefficientwise HK theorem). *For every $m \geq 2$ and every $n \geq 0$,*

$$L_n(x) \geq_c R_n(x).$$

Equivalently, $D_n(x) \in \mathbb{N}[x]$ for all $n \geq 0$.

Proof. The case $n = 0$ is equality. For $n \geq 1$, we prove simultaneously that

$$D_n \geq_c 0, \tag{5}$$

and the stronger odd-depth assertion

$$D_n \geq_c x F_{n-2} A_n \quad (n \geq 3 \text{ odd}). \tag{6}$$

The ordinary base cases are

$$D_1 = L_1 - R_1 = x(F_0^{m-1} - 1) \geq_c 0,$$

and

$$\begin{aligned} D_2 &= x F_0^{m-1} F_1^{m-1} + x^2 F_0^{m^2-1} - x F_0^{m^2} \\ &= x F_0^{m-1} (F_1^{m-1} - F_0^{m(m-1)}) \geq_c 0, \end{aligned}$$

because $F_1 = F_0^m + x \geq_c F_0^m$.

We also need the strong base case $n = 3$. By [Lemma 4.3](#),

$$D_3 = F_2^{m-1} D_2 + x F_1^{m^2-1} D_1 + x F_0^m A_3.$$

Therefore

$$\begin{aligned} D_3 - xF_1A_3 &= F_2^{m-1}D_2 + xF_1^{m^2-1}D_1 + x(F_0^m - F_1)A_3 \\ &= F_2^{m-1}D_2 + x^2 \left[(F_0^{m-1} - 1)F_1^{m^2-1} - A_3 \right]. \end{aligned}$$

The first term is non-negative. For the bracket, using the definition of A_3 ,

$$\begin{aligned} (F_0^{m-1} - 1)F_1^{m^2-1} - A_3 &= F_0^{m-1}F_1^{m^2-1} - F_0^{m(m-1)}F_2^{m-1} \\ &= (F_0F_1^{m+1})^{m-1} - (F_0^mF_2)^{m-1}. \end{aligned}$$

It remains to show $F_0F_1^{m+1} \geq_c F_0^mF_2$, equivalently $F_1^{m+1} \geq_c F_0^{m-1}F_2$. Since $F_2 = F_1^m + xF_0^{m^2}$,

$$\begin{aligned} F_1^{m+1} - F_0^{m-1}F_2 &= F_1^m(F_1 - F_0^{m-1}) - xF_0^{m^2+m-1} \\ &= x \left[F_1^m(F_0^{m-1} + 1) - F_0^{m^2+m-1} \right] \\ &\geq_c x \left[F_0^{m^2}(F_0^{m-1} + 1) - F_0^{m^2+m-1} \right] \\ &= xF_0^{m^2} \geq_c 0. \end{aligned}$$

Thus $D_3 \geq_c xF_1A_3$, establishing the strong base case.

Now let $n \geq 4$ and assume the relevant assertions for smaller depths. First suppose that n is even. Then $n-1$ is odd, so the strong induction hypothesis gives

$$D_{n-1} \geq_c xF_{n-3}A_{n-1},$$

and the ordinary induction hypothesis gives $D_{n-2} \geq_c 0$. By [Lemma 4.3](#),

$$\begin{aligned} D_n &= F_{n-1}^{m-1}D_{n-1} + xF_{n-2}^{m^2-1}D_{n-2} + xF_{n-3}^m A_n \\ &\geq_c xF_{n-3}F_{n-1}^{m-1}A_{n-1} + xF_{n-3}^m A_n \\ &= xF_{n-3} (F_{n-1}^{m-1}A_{n-1} + F_{n-3}^{m-1}A_n) \geq_c 0, \end{aligned}$$

where the final inequality is [Lemma 4.4](#). This proves the ordinary assertion for even n .

Now suppose that $n \geq 5$ is odd. We prove the stronger assertion. Using [Lemma 4.3](#),

$$\begin{aligned} D_n - xF_{n-2}A_n &= F_{n-1}^{m-1}D_{n-1} + xF_{n-2}^{m^2-1}D_{n-2} + x(F_{n-3}^m - F_{n-2})A_n \\ &= F_{n-1}^{m-1}D_{n-1} + xF_{n-2}^{m^2-1}D_{n-2} - x^2F_{n-4}^{m^2}A_n. \end{aligned}$$

By ordinary induction, $D_{n-1} \geq_c 0$. Since $n-2$ is odd, the strong induction hypothesis gives

$$D_{n-2} \geq_c xF_{n-4}A_{n-2}.$$

Therefore

$$D_n - xF_{n-2}A_n \geq_c x^2F_{n-4} \left(F_{n-2}^{m^2-1}A_{n-2} - F_{n-4}^{m^2-1}A_n \right) \geq_c 0,$$

where the final inequality is [Lemma 4.5](#). This proves the strong assertion for odd n , and hence also the ordinary assertion. The simultaneous induction is complete. $\square$

Corollary 4.7 (HK-property for perfect m -ary trees). *Every perfect m -ary tree has the HK-property.*

Proof. By [Theorem 4.6](#), every perfect m -ary tree satisfies leaf-root domination at every depth. The result follows from [Lemma 2.2](#). $\square$

5 Quantifying the HK gap

The proof above already contains a recurrence for the gap polynomial $D_n = L_n - R_n$. We now state it as the structural refinement of the HK-property.

Theorem 5.1 (HK gap decomposition). *For every $n \geq 3$,*

$$D_n = F_{n-1}^{m-1} D_{n-1} + x F_{n-2}^{m^2-1} D_{n-2} + \Gamma_n,$$

where

$$\Gamma_n = x F_{n-3}^m \left(F_{n-3}^{m(m-1)} F_{n-1}^{m-1} - F_{n-2}^{m^2-1} \right).$$

Moreover,

$$\Gamma_n \geq_c 0 \quad \text{for odd } n \geq 3,$$

and

$$\Gamma_n \leq_c 0 \quad \text{for even } n \geq 4.$$

Proof. The recurrence is [Lemma 4.3](#), with $\Gamma_n = x F_{n-3}^m A_n$. Since $x F_{n-3}^m \in \mathbb{N}[x]$, the sign of Γ_n is the sign of A_n . The parity statement follows from [Lemma 4.1](#). $\square$

Remark 5.2. The gap decomposition is not logically needed after [Theorem 4.6](#); it is a refinement of it. The theorem says that the difference between the leaf star and root star is governed by a second-order inhomogeneous recurrence. The forcing term Γ_n changes sign with the depth, so the non-negativity of D_n is not merely a term-by-term consequence of the decomposition.

For fixed n , write

$$D_n(x) = \sum_{k \geq 0} \delta_{n,k} x^k.$$

Then $\delta_{n,k}$ is the exact difference

$$\delta_{n,k} = |\mathcal{I}_\ell^k(T_{m,n})| - |\mathcal{I}_r^k(T_{m,n})|.$$

Thus [Theorem 5.1](#) gives a recursive way to study the full coefficient sequence of the HK gap.

6 Partial k -EKR classification

We collect the k -EKR consequences that are independent of the coefficientwise HK proof. The first result gives a general positive range for all trees.

Theorem 6.1 (A general positive range for trees). *Let T be a tree on N vertices. If $k \geq 3$ and $N \geq 2k^2$, then T is strictly k -EKR. Consequently,*

$$3 \leq k \leq \sqrt{\frac{N_{m,n}}{2}} \quad \implies \quad T_{m,n} \text{ is strictly } k\text{-EKR}.$$

In this range, [Corollary 4.7](#) identifies the extremal stars in $T_{m,n}$ as leaf-stars.

Theorem 6.2 (Complete 2-EKR classification for trees). *Let T be a tree on $N \geq 3$ vertices. Then*

$$T \text{ is not 2-EKR} \iff T \cong K_{1,3}.$$

Moreover,

$$T \text{ is 2-EKR but not strictly 2-EKR} \iff N = 5.$$

Thus, for perfect m -ary trees with $n \geq 1$,

$$T_{m,n} \text{ is not 2-EKR} \iff (m, n) = (3, 1),$$

and

$$T_{m,n} \text{ is 2-EKR but not strictly 2-EKR} \iff (m, n) = (4, 1).$$

All other perfect m -ary trees with independent 2-sets are strictly 2-EKR.

Theorem 6.3 (Complete 3-EKR classification for perfect m -ary trees). *Let $T_{m,n}$ be a perfect m -ary tree with $\alpha_{m,n} \geq 3$. Then*

$$T_{m,n} \text{ is not 3-EKR} \iff (n = 1, 4 \leq m \leq 5) \text{ or } (m, n) = (2, 2).$$

Moreover,

$$T_{m,n} \text{ is 3-EKR but not strictly 3-EKR} \iff (m, n) = (6, 1).$$

All remaining cases with independent triples are strictly 3-EKR.

Theorem 6.4 (Upper-layer failures). *Let G be a graph on N vertices with independence number $\alpha(G)$. If*

$$\frac{N}{2} < k < \alpha(G),$$

then G is not k -EKR.

For perfect m -ary trees, the following sharper top-layer statements hold.

(i) *If $m \geq 3$, $n \geq 2$, and $1 \leq t \leq m - 1$, then*

$$T_{m,n} \text{ is not } (\alpha_{m,n} - t)\text{-EKR}.$$

(ii) *If $m \geq 2$, $n \geq 1$, and $\alpha_{m,n} \geq 3$, then*

$$T_{m,n} \text{ is not } (\alpha_{m,n} - 1)\text{-EKR}.$$

The proofs of these statements are given in [Appendix A](#).

7 Open problems

The generating-function framework raises several natural questions.

Question 7.1. Is the coefficient sequence of $D_n(x) = L_n(x) - R_n(x)$ unimodal for every $m \geq 2$ and every $n \geq 0$?

Question 7.2. Is $D_n(x)$ log-concave, or real-rooted, for any infinite family of pairs (m, n) ?

Question 7.3. Determine asymptotics for

$$\max_k [x^k] D_n(x)$$

as $n \rightarrow \infty$ with m fixed.

Question 7.4. Give a complete k -EKR classification for perfect m -ary trees. The low-uniformity and upper-layer results above leave a large intermediate range.

Question 7.5. Can the generating-function framework be extended to non-perfect rooted trees, or to broader families of trees for which the HK-property is known or conjectured?

A Proofs for further k -EKR phenomena

Lemma A.1 (Hilton–Milner transfer). *Let G be a graph on N vertices, and let*

$$S_k(G) = \max_{v \in V(G)} |\mathcal{I}_v^k(G)|.$$

If $k \geq 3$ and

$$S_k(G) > \binom{N-1}{k-1} - \binom{N-k-1}{k-1} + 1,$$

then G is strictly k -EKR.

Proof. Let $\mathcal{F} \subseteq \mathcal{I}^k(G)$ be intersecting. If $\bigcap_{F \in \mathcal{F}} F \neq \emptyset$, then $\mathcal{F}$ is contained in a star, so $|\mathcal{F}| \leq S_k(G)$. Otherwise $\mathcal{F}$ is a nontrivial intersecting family of ordinary k -subsets of an N -element set. By the Hilton–Milner theorem [3],

$$|\mathcal{F}| \leq \binom{N-1}{k-1} - \binom{N-k-1}{k-1} + 1 < S_k(G).$$

Thus every maximum intersecting family is a star. □

Proof of Theorem 6.1. Let ℓ be a leaf of T and let p be its parent. The k -star centred at ℓ is in bijection with independent $(k-1)$ -sets of $T - \{\ell, p\}$. This graph is a forest on $N-2$ vertices with at most $N-3$ edges. Hence

$$S_k(T) \geq \binom{N-2}{k-1} - (N-3) \binom{N-4}{k-3},$$

by subtracting, for each edge, the $(k-1)$ -sets containing that edge. Put $B = \binom{N-2}{k-2}$. Then

$$\binom{N-2}{k-1} = \frac{N-k}{k-1} B$$

and

$$(N-3) \binom{N-4}{k-3} = \frac{(k-2)(N-k)}{N-2} B.$$

Therefore

$$S_k(T) \geq B \cdot \frac{(N-k)(N-2-(k-1)(k-2))}{(k-1)(N-2)}.$$

If $N \geq 2k^2$ and $k \geq 3$, the factor on the right is larger than $k+1$. Indeed, after clearing denominators this is equivalent to positivity of

$$D(N, k) = N^2 - 2Nk^2 + 2Nk - 3N + k^3 - k^2 + 4k - 2.$$

For fixed $k \geq 3$, this polynomial is increasing for $N \geq 2k^2$, and

$$D(2k^2, k) = 5k^3 - 7k^2 + 4k - 2 > 0.$$

Thus $S_k(T) > (k+1)B > kB + 1$. Finally,

$$\binom{N-1}{k-1} - \binom{N-k-1}{k-1} + 1 \leq k \binom{N-2}{k-2} + 1 = kB + 1,$$

by a union bound over the k possible elements of a fixed k -set that another k -set may meet. The result follows from Lemma A.1. The statement for $T_{m,n}$ follows from $N = N_{m,n}$. □

Proof of Theorem 6.2. A pairwise intersecting family of 2-sets is either contained in a star or consists of the three 2-subsets of a 3-set. For a tree T on N vertices, the independent 2-star centred at v has size $N - 1 - \deg(v)$, so the largest independent 2-star has size $N - 2$ because every nontrivial tree has a leaf.

If $N \geq 6$, then $N - 2 \geq 4$, while every non-star intersecting 2-family has size at most 3; hence T is strictly 2-EKR. If $N = 5$, then the largest star has size 3, and every tree on five vertices has an independent triple, whose three pairs form a non-star intersecting family of size 3. Hence such trees are 2-EKR but not strictly so. If $N = 4$, the only trees are P_4 and $K_{1,3}$. The path P_4 has no independent triple and is strictly 2-EKR, while in $K_{1,3}$ the three leaf-pairs form an intersecting family of size 3, larger than the largest star of size 2. The case $N = 3$ is trivial.

For perfect m -ary trees, $T_{m,1} = K_{1,m}$. Thus the failure case is $(m, n) = (3, 1)$ and the non-strict case is $(m, n) = (4, 1)$. All cases with $n \geq 2$ have at least seven vertices and are strictly 2-EKR. $\square$

Lemma A.2. *Every tree on $N \geq 11$ vertices is strictly 3-EKR.*

Proof. Let ℓ be a leaf with parent p . The 3-star centred at ℓ is in bijection with independent 2-sets of $T - \{\ell, p\}$. This forest has $N - 2$ vertices and at most $N - 3$ edges, so the star has size at least

$$\binom{N-2}{2} - (N-3) = \frac{(N-3)(N-4)}{2}.$$

If $\mathcal{F} \subseteq \mathcal{I}^3(T)$ is intersecting with empty total intersection, the Hilton–Milner theorem [3] gives $|\mathcal{F}| \leq 3N - 8$. For $N \geq 11$,

$$\frac{(N-3)(N-4)}{2} > 3N - 8.$$

Therefore every maximum intersecting family is a star. $\square$

Proof of Theorem 6.3. If $n \geq 3$, then $|V(T_{m,n})| \geq |V(T_{2,3})| = 15$, so Lemma A.2 gives strict 3-EKR. If $n = 2$, then $|V(T_{m,2})| = 1 + m + m^2$. For $m \geq 3$ this is at least 13, so again the tree is strictly 3-EKR. It remains to handle $T_{2,2}$.

In $T_{2,2}$, let r be the root and let L be the set of four leaves. Then $A = \{r\} \cup L$ is an independent set of size 5, so $\binom{A}{3}$ is an intersecting family of size 10. A leaf-star has size 7: after fixing a leaf, the remaining two vertices must be chosen independently from the five vertices outside that leaf and its parent, and exactly three forbidden pairs remain. The root-star has size $\binom{4}{2} = 6$, and level-one stars are smaller. Hence $T_{2,2}$ is not 3-EKR.

Finally let $n = 1$. Then $T_{m,1} = K_{1,m}$, so independent 3-sets are exactly the 3-subsets of the m leaves. The problem reduces to the ordinary 3-uniform EKR theorem on an m -element ground set. For $m = 3$, there is a unique independent triple, so the tree is strictly 3-EKR. For $m = 4, 5$, the whole layer of triples is intersecting and larger than a star, so the tree is not 3-EKR. For $m = 6$, ordinary EKR gives maximum size $\binom{5}{2} = 10$, and non-star maximum families exist by choosing one triple from each complementary pair. Hence the tree is 3-EKR but not strictly so. For $m \geq 7$, the strict form of the ordinary EKR theorem gives that the only maximum families are stars. $\square$

Lemma A.3 (Large layers are automatically intersecting). *Let G be a graph on N vertices with independence number $\alpha(G)$. If $N/2 < k < \alpha(G)$, then G is not k -EKR.*

Proof. Since $2k > N$, any two k -subsets of $V(G)$ intersect. Hence the whole layer $\mathcal{I}^k(G)$ is an intersecting family. It is not a star: if A is an independent set of size $\alpha(G)$, then for every vertex v there exists an independent k -subset of A avoiding v ; if $v \in A$, use $k < \alpha(G)$. Thus no vertex lies in every member of $\mathcal{I}^k(G)$. $\square$

Lemma A.4 (Expansion from the opposite parity class). *Let $m \geq 3$ and $n \geq 2$. Let*

$$A = \{\text{vertices of } T_{m,n} \text{ at depths } n, n-2, n-4, \dots\}$$

and $B = V(T_{m,n}) \setminus A$. If $\emptyset \neq Y \subseteq B$, then

$$|N(Y)| - |Y| \geq m - 1.$$

Proof. Consider the bipartite forest induced by $Y \cup N(Y)$. Let it have c connected components and e edges. Then

$$e = |Y| + |N(Y)| - c.$$

Every vertex of B has at least m neighbours in A : the root has m children when it lies in B , and every other vertex of B has its parent and its m children in A . Therefore $e \geq m|Y|$. Hence

$$|N(Y)| = e - |Y| + c \geq (m-1)|Y| + c,$$

and so

$$|N(Y)| - |Y| \geq (m-2)|Y| + c \geq m - 1.$$

□

Lemma A.5 (Penultimate expansion). *Let $T_{m,n}$ be a perfect m -ary tree, and let A be the maximum independent set consisting of depths $n, n-2, n-4, \dots$. Put $B = V(T_{m,n}) \setminus A$. If $\emptyset \neq Y \subseteq B$, then*

$$|N(Y)| \geq |Y| + 1.$$

Moreover, equality can occur only when $m = 2$, n is odd, and Y is the singleton consisting of the root.

Proof. Use the same forest-edge count. If $m \geq 3$, every vertex of Y has at least three neighbours in A , so $e \geq 3|Y|$ and hence

$$|N(Y)| = e - |Y| + c \geq 2|Y| + c \geq |Y| + 2.$$

Now suppose $m = 2$. Every non-root vertex of B has three neighbours in A , while the root has two neighbours in A when it belongs to B , which occurs exactly when n is odd. Let $\rho = 1$ if the root belongs to Y , and $\rho = 0$ otherwise. Then $e \geq 3|Y| - \rho$, so

$$|N(Y)| \geq 2|Y| - \rho + c.$$

If $\rho = 0$, this is at least $|Y| + 2$. If $\rho = 1$ and $|Y| = 1$, then Y is the root singleton and equality holds. If $\rho = 1$ and $|Y| \geq 2$, then $|N(Y)| \geq 2|Y| \geq |Y| + 2$. □

Proof of Theorem 6.4. The first assertion is Lemma A.3.

For the top-block statement, assume $m \geq 3$ and $n \geq 2$. Let

$$A = \{\text{vertices at depths } n, n-2, n-4, \dots\}, \quad B = V(T_{m,n}) \setminus A.$$

Then A is a maximum independent set of size $\alpha_{m,n}$. Let $I \in \mathcal{I}^{\alpha_{m,n}-t}(T_{m,n})$ and set $Y = I \cap B$. Since the vertices selected from A must avoid $N(Y)$,

$$|I| \leq |Y| + \alpha_{m,n} - |N(Y)| = \alpha_{m,n} - (|N(Y)| - |Y|).$$

If $Y \neq \emptyset$, [Lemma A.4](#) gives

$$|I| \leq \alpha_{m,n} - (m - 1).$$

Thus, if $t < m - 1$, every independent $(\alpha_{m,n} - t)$ -set is contained in A .

At the endpoint $t = m - 1$, equality can occur only when $|N(Y)| - |Y| = m - 1$. Inspecting the proof of [Lemma A.4](#), equality forces Y to be a singleton and forces its unique vertex to have exactly m neighbours in A . In $T_{m,n}$ this can only be the root, and only when the root belongs to B , i.e. when n is odd. Hence the independent $(\alpha_{m,n} - (m - 1))$ -sets are all subsets of A of that size, together with, in this exceptional case, the single additional set

$$E = \{r\} \cup (A \setminus N(r)).$$

Since $n \geq 2$, we have $\alpha_{m,n} \geq m^2 + 1$. Therefore, for $1 \leq t \leq m - 1$,

$$\alpha_{m,n} - t > \frac{\alpha_{m,n}}{2}.$$

So any two $(\alpha_{m,n} - t)$ -subsets of A intersect. In the exceptional endpoint case, the set E also intersects every $(\alpha_{m,n} - (m - 1))$ -subset of A , because such a subset omits only $m - 1$ vertices of A while $N(r)$ has m vertices. Thus the whole layer is intersecting.

This layer is not a star. For every vertex v , there is a member of the layer avoiding v : if $v \in A$, choose an $(\alpha_{m,n} - t)$ -subset of $A \setminus \{v\}$; if $v \in B$, choose an $(\alpha_{m,n} - t)$ -subset of A . Therefore the whole layer is an intersecting family strictly larger than every star, and $T_{m,n}$ is not $(\alpha_{m,n} - t)$ -EKR.

For the penultimate statement, let $m \geq 2$, $n \geq 1$, and $\alpha_{m,n} \geq 3$. Again put $A = \{\text{depths } n, n - 2, n - 4, \dots\}$ and $B = V(T_{m,n}) \setminus A$. If $I \in \mathcal{I}^{\alpha_{m,n}-1}(T_{m,n})$ and $Y = I \cap B$, then

$$|I| \leq |Y| + \alpha_{m,n} - |N(Y)|.$$

By [Lemma A.5](#), every independent $(\alpha_{m,n} - 1)$ -set is either of the form $A \setminus \{a\}$, or, in the exceptional case $m = 2$ and n is odd, is possibly also the set

$$E = \{r\} \cup (A \setminus N(r)).$$

The sets $A \setminus \{a\}$ are pairwise intersecting because $\alpha_{m,n} \geq 3$. In the exceptional case, E also intersects every $A \setminus \{a\}$: when $m = 2$ and n is odd with $\alpha_{m,n} \geq 3$, the set $A \setminus N(r)$ is nonempty, and $A \setminus \{a\}$ cannot delete all of it. Hence the entire $(\alpha_{m,n} - 1)$ -layer is intersecting.

The layer is not a star, since each vertex is avoided by some member of the layer. Thus $T_{m,n}$ is not $(\alpha_{m,n} - 1)$ -EKR. $\square$

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